NBTI Performance Enhancement with Process Integration of High Current Fluorine Incorporation and O_2 Gas Asher Process in 45nm CMOS Technology

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Index Terms- Negative bias temperature instability (NBTI), High Current Fluorine (HC F), Source/Drain extension (S/D extension), Post Poly Resist Strip (PRS), Spacer Etch (SPE), Saturation Drain Current (Id_{sat}).

ABSTRACT

In common agreement with Negative Bias Temperature instability (NBTI) as a serious Front-end reliability issue, great efforts were made in recent years to investigate NBTI mechanism, characterization techniques and performance improvement. In this work authors focused on device NBTI performance improvement from combined impact of multiple process steps integration relating to SiON/Si interface quality. The effect of process integration of High Current Fluorine incorporation in Source/Drain extension along with O_2 gas Asher process for post poly resist strip (PRS) and spacer etch process (SPE) on NBTI performance for thin gate regular V_t transistor and SRAM were investigated. Through our work, we demonstrated that by correctly chosen thermal budget and nitrogen profile, the high current fluorine implantation in S/D extension and O_2 gas asher process decrease interface state component of NBTI to greater extent and thus facilitate to achieve enhanced NBTI performance to meet end-of-lifetime specifications.

I. INTRODUCTION

Researchers around the world are in common agreement that NBTI induced PMOS device instability (shift in saturation drain current, threshold voltage, transconductance, etc.,) is a major reliability concern. With this a lot of efforts have been made to understand as well as to address the issue from process, doping species, test methodology, modelling and process integration perspective [1] [2]. In recent years, major works have been done in NBTI performance improvement with fluorine incorporation into Source/Drain extension and silicon substrate. Da-Yuan Lee et al. reported NBTI effect reduced effectively by BF2+ implant compared to counterpart with boron in source/drain implantation [1]. C.C. Chen et al., and T.B Hook et al., reported that thermal budget must be carefully chosen to avoid the occurrence of Boron penetration, so that with proper fluorine incorporation interfacial layer strength can be enhanced [3] [4]. Masayuki Terai et al., reported that the fluorine incorporation decrease the interface component in plasma-nitride SiON gates, while it does not decrease the bulk component [5].

In our previous work, we had reported that O_2+N_2 ashing in poly resist strip and spacer etch asher process shows improved NBTI performance in terms of saturation drain current degradation and spread compared to H_2+N_2 ashing as shown in Fig.1. This was due to less plasma damage to gate oxide under poly edge coverage because of shorter RF time of O_2+N_2 plasma [6]. Less charging damage during processing creates smaller number of interface states, which attributes to lower number of Si-H bonds [6] [7].

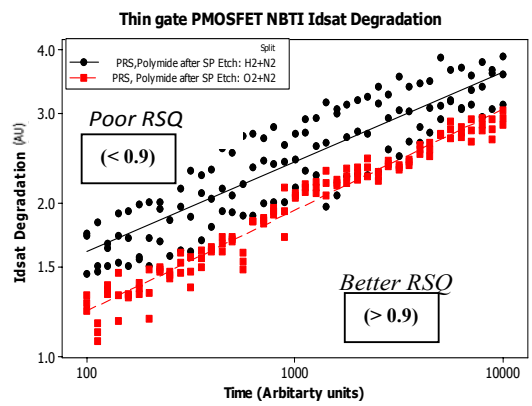

Fig.1. Thin gate pMOSFET NBTI Id_{sat} Degradation plot (O_2+N_2 Gas Ash Process Vs H_2+N_2 Gas Process for PRS & Polymer removal after SP Etch)

In present work team had put efforts to integrate different process steps which will results in improvement of NBTI performance. The effect of process integration of additional high current fluorine incorporation in source/drain extension along with O_2 gas asher process in PRS and SPE on NBTI performance was investigated. Thermal budget was carefully chosen to avoid boron penetration and nitrogen profile was properly controlled to reduce number of nitrogen atoms at SiON/Si interface.

II. EXPERIMENTS

Thin gate p-MOSFET Samples with P+ Poly-SiON/Si stacks are fabricated using 45nm low power CMOS process. The gate oxide thickness is about 16A. High current Fluorine was introduced into Source/Drain by the implant system. The process flow illustration of HC F implantation is as shown in Fig.2. Offset spacer is formed after poly module [Fig.2(a)] followed by Source/Drain extension BF_2 implant. Additional high current fluorine implant was then implanted

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into Source/Drain extension on both sides of the poly-Si electrode [Fig.2(b)]. Fluorine was then diffused to the SiON/Si-substrate interface by subsequent annealing activation with temperature up to 1000 ◦C [Fig.2(c)]. NBTI experiments were conducted on regular V_t transistor and SRAM. The split information is as shown in Table.1 and Table. 2. On-the-fly characterization technique was used to characterize NBTI in our experiments. Devices practically observe no relaxation, as the negative gate bias is never removed during stress and measurement [8]. The shift in saturation drain current (Id_{sat}) was monitored.

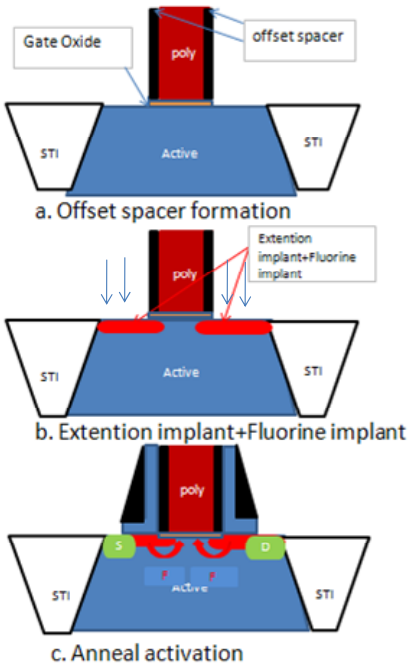

Fig.2. Process flow for Fluorine Incorporation

Thin Gate PMOSFET	Post Poly Etch	Spacer Etch Asher	Source/Drain Extension Implant
Split A	O2+N2	O2+N2	BF2 Implant
Split B	O2+N2	O2+N2	BF2 Implant + High Current Fluorine Implant

Table.1. Thin Gate PMOSFET Process Split Information

SRAM PMOSFET	Post Poly Etch	Spacer Etch Asher	Source/Drain Extension Implant
Split A	H2+N2	H2+N2	BF2 Implant
Split B	O2+N2	O2+N2	BF2 Implant + High Current Fluorine Implant

Table.2. SRAM PMOSFET Process Split Information

III. RESULTS AND DISCUSSION

A. Impact of Fluorine Incorporation on Saturation Drain current for Thin Gate Regular V_t transistor.

Fig.3 shows NBTI induced saturation drain current degradation comparison plot for the samples that were fabricated by BF_2 Implant in S/D extension (Split A) and with additional high current fluorine incorporation in source/drain extension (Split B). Both the splits use O_2+N_2 gas in post poly resist strip and spacer etch asher process. The process split table is as shown in Table.1.

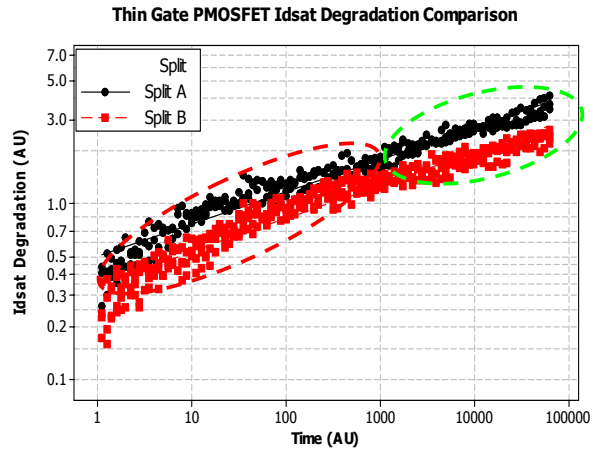

Fig.3. Thin gate pMOSFET NBTI Id_{sat} Degradation plot (BF_2 Implant + O_2 Gas Asher Process Vs HC Fluorine + O_2 Gas Asher Process)

Split B with additional high current fluorine implant in S/D extension shows enhanced NBTI performance in terms of saturation drain current degradation compared to Split A. At lower stress time (highlighted in red circle) both split A and Split B shows approximately same amount of Id_{sat} degradation but at higher stress time (highlighted in green circle) split B shows lower degradation compared to Split A. This result indicates that the additional fluorine atoms that were implanted in source/drain extension diffuse sufficiently towards SiON/Si interface by subsequent annealing steps and thus by reducing the number of interface states that attributes to lower number of Si-H dangling bonds. The binding energy of silicon dangling bonds and fluorine atoms (540 KJ/mol) is higher than that of silicon dangling bonds with hydrogen(298 KJ/mol) [9]. At lower stress time for split B thin gate regular V_t transistor, the time evolution slope is around 0.19 and is around 0.145 at higher stress time. Lowering in time evolution slope at higher stress time indicates a tendency of device saturation.

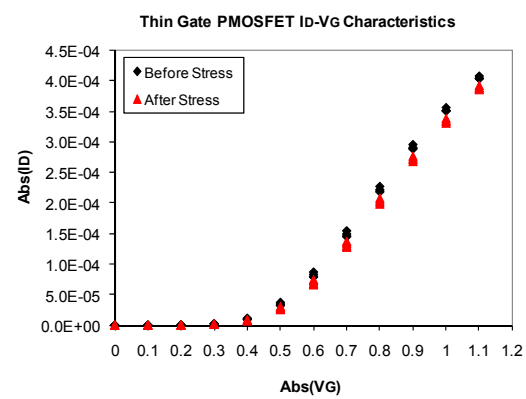

Fig.4. Thin gate pMOSFET Initial and Stressed ID-VG curve for samples fabricated with High Current Fluorine S/D extension Implant and O_2 Gas Asher Process

Fig.4 shows pre and post stress ID-VG characteristics for Split B.

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The threshold voltage was shifted by 11.3mV from initial after NBTI stress. Fig.5 shows the Id_{sat} degradation comparison for both the splits at certain end of lifetime target. At the end-of-lifetime target split B shows approximately 4% lower Id_{sat} degradation compared to split A.

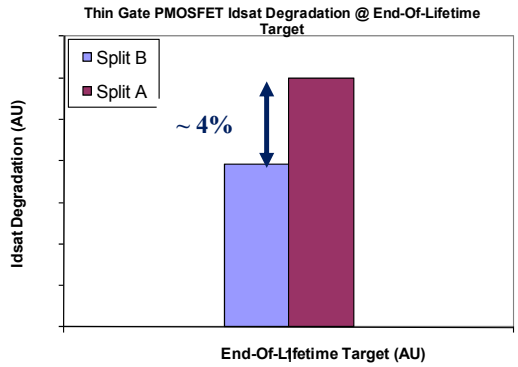

Fig.5. Thin gate pMOSFET Id_{sat} degradation @ target lifetime comparison (BF_2 Implant + O_2 Gas Asher Process Vs HC Fluorine + O_2 Gas Asher Process)

B. Impact of Fluorine Incorporation on Saturation Drain current for SRAM.

Fig.6 shows NBTI induced saturation drain current degradation comparison plot for the samples that were fabricated with S/D extension BF_2 Implant (Split A) and additional high current Fluorine implant (Split B) in source/drain extension.

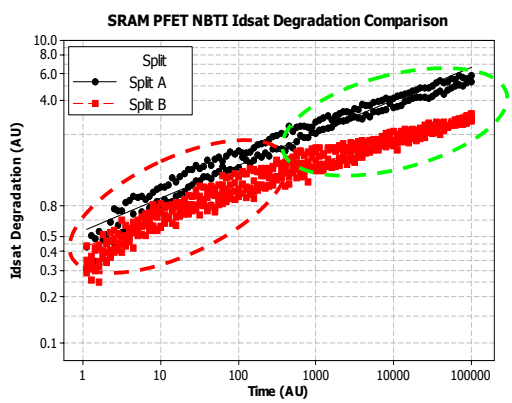

Fig.6. SRAM pMOSFET NBTI Id_{sat} Degradation plot (BF_2 Implant + H_2 Gas Asher Process Vs HC Fluorine + O_2 Gas Asher Process)

With the NBTI enhancement observed in thin gate regular V_t transistor with O_2 gas asher process we fabricated and tested devices with O_2 gas asher process integrated with BF_2 implant plus additional high current fluorine incorporation in S/D extension (Split B) and NBTI performance is compared with split A. SRAM process split table is as shown in Table. 2. In case of SRAM, Split B (S/D extension BF_2 implant plus additional high current fluorine implant) also shows enhanced NBTI performance in terms of saturation drain current degradation compared to Split A (with only S/D extension BF_2 implant). This is also because of the reduce number of interface states at SiON/Si interface.

In Fig.6 at lower stress time both the splits shows the same amount of Id_{sat} degradation (highlighted in red circle) and at higher stress time split B shows much lower Id_{sat} degradation (highlighted in green circle) compared to Split A indicating lower number of interface states at SiON/Si interface because of combined effect of fluorine incorporation and O_2 gas asher process. At lower stress time, the time evolution slope for HDSRAM split B is around 0.18 and at higher stress time is around 0.137. Similar to regular V_t transistor, HDSRAM also shows fractional time dependence of NBTI degradation slowly decreases at longer stress time and indicates a tendency towards device saturation. Fig.7 shows pre and post stress ID-VG characteristics for Split B. The threshold voltage was shifted by 20mV from initial after NBTI stress.

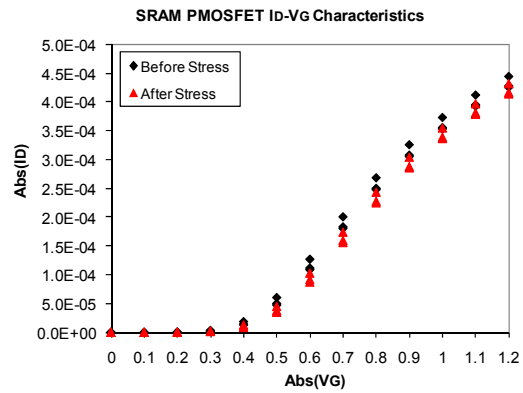

Fig.7. SRAM pMOSFET Initial and Stressed ID-VG curve for samples fabricated with BF_2 Implant + High Current Fluorine S/D extension Implant and O_2 Gas Asher Process

Fig.8 shows the Id_{sat} degradation comparison for 2 splits at end of lifetime target. At the end-of-lifetime target split B shows approximately 10% lower Id_{sat} degradation compared to split A.

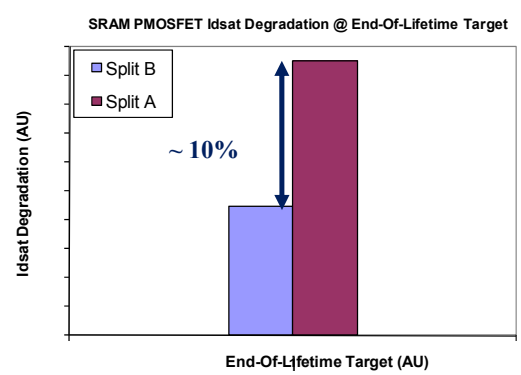

Fig.8. SRAM pMOSFET Id_{sat} degradation @ target lifetime comparison (BF_2 Implant + H_2 Gas Asher Process Vs HC Fluorine + O_2 Gas Asher Process)

From thin gate regular V_t transistor and SRAM end of lifetime results we can approximate that 2 to 3 % of NBTI improvement is contributed from O_2 gas asher process (instead of H_2 gas asher process) integrated with S/D extension BF_2 implant and additional 4 to 5% of NBTI improvement with additional High current Fluorine implant at S/D extension. The NBTI reliability performance is as shown in Table. 3.

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Post Poly Etch	Spacer Etch Asher	Source/Drain Extension Implant	Reliability performance
H2+N2	H2+N2	BF2 Implant	Poor NBTI
O2+N2	O2+N2	BF2 Implant	Improved NBTI
O2+N2	O2+N2	BF2 Implant + HC Fluorine Implant	Enhanced NBTI

Table.3. Reliability NBTI performance comparison for different post poly resist stripping (PRS), Polymer removal after spacer etch process and LDD Implant

IV. Summary

The impact of process integration of high current fluorine implant in Source/Drain extension with O_2+N_2 gas asher process for PRS and SPE on NBTI performance were investigated. Integration of both process (O_2 gas asher process and HC Fluorine implant) enhance NBTI performance. O_2 Gas asher process improves NBTI performance because of less plasma damage to gate oxide under poly edge coverage because of the shorter RF time of O_2+N_2 plasma. Additional high current fluorine incorporation proved to further enhance NBTI performance by decreasing the number of interface states and thus meet stringent reliability end of lifetime specifications with good margin.

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